Table 1: L2 (TDOST classes), all classes — comparable to TDOST / DeepCASAS; $m = 5$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.70	0.38	0.88	0.50	0.73	0.29	0.77	0.39
k-means	Sensor+ToD hist.	0.71	0.28	0.78	0.32	0.84	0.39	0.78	0.33
k-means	MLM	0.77	0.33	**0.89**	**0.51**	0.74	0.29	0.80	0.38
k-means	MLM-time	0.73	0.32	0.89	0.50	0.77	0.32	0.79	0.38
k-means	MLM + ToD	0.72	0.29	0.81	0.32	**0.86**	**0.44**	0.80	0.35
k-means	MLM-time + ToD	0.74	0.32	0.76	0.31	**0.86**	0.41	0.79	0.35
SCAN	MLM	0.76	0.33	0.85	0.36	0.77	0.33	0.79	0.34
SCAN	MLM-time	0.75	0.39	0.88	0.49	0.77	0.34	0.80	**0.41**
DEC	MLM	**0.78**	0.38	0.89	0.47	0.77	0.32	**0.81**	0.39
DEC	MLM-time	**0.78**	**0.40**	**0.89**	0.47	0.78	0.35	**0.82**	**0.41**
$K = 100$ *clusters*									
k-means	Sensor hist.	0.78	0.38	**0.89**	0.50	0.77	0.38	0.81	0.42
k-means	Sensor+ToD hist.	0.79	0.43	0.87	0.47	**0.87**	**0.50**	0.85	0.47
k-means	MLM	0.80	0.46	**0.89**	0.49	0.80	0.44	0.83	0.46
k-means	MLM-time	0.79	0.46	0.89	0.49	0.79	0.35	0.82	0.43
k-means	MLM + ToD	**0.81**	0.43	0.89	0.48	**0.88**	0.48	**0.86**	0.46
k-means	MLM-time + ToD	0.80	0.43	0.88	0.51	**0.87**	0.48	0.85	0.48
SCAN	MLM	0.80	**0.47**	0.88	**0.58**	0.81	0.41	0.83	**0.48**
SCAN	MLM-time	0.79	0.41	0.85	0.55	0.81	0.46	0.82	0.47
DEC	MLM	0.78	0.40	0.88	0.46	0.79	0.35	0.82	0.41
DEC	MLM-time	0.79	0.41	**0.89**	0.51	0.79	0.35	0.82	0.42

Table 2: L2, rare classes masked from the average; $m = 5$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.71	0.45	0.88	0.79	0.73	0.33	0.77	0.53
k-means	Sensor+ToD hist.	0.71	0.35	0.78	0.50	0.84	0.46	0.78	0.44
k-means	MLM	0.77	0.42	**0.90**	**0.81**	0.75	0.34	0.80	0.52
k-means	MLM-time	0.73	0.39	0.89	0.80	0.77	0.37	0.80	0.52
k-means	MLM + ToD	0.73	0.36	0.81	0.52	**0.86**	**0.52**	0.80	0.46
k-means	MLM-time + ToD	0.74	0.40	0.76	0.49	**0.86**	0.48	0.79	0.46
SCAN	MLM	0.77	0.41	0.85	0.58	0.77	0.39	0.80	0.46
SCAN	MLM-time	0.76	0.49	0.88	0.78	0.77	0.40	0.80	**0.56**
DEC	MLM	**0.79**	0.47	0.89	0.75	0.77	0.37	**0.81**	0.53
DEC	MLM-time	**0.79**	**0.50**	**0.89**	0.75	0.78	0.41	**0.82**	**0.55**
$K = 100$ *clusters*									
k-means	Sensor hist.	0.78	0.48	**0.89**	**0.80**	0.77	0.38	0.81	0.56
k-means	Sensor+ToD hist.	0.79	0.52	0.88	0.76	**0.88**	0.52	0.85	0.60
k-means	MLM	0.80	0.57	**0.89**	0.79	0.80	0.43	0.83	0.60
k-means	MLM-time	0.80	0.57	0.89	0.79	0.79	0.41	0.82	0.59
k-means	MLM + ToD	**0.82**	0.53	0.89	0.77	**0.88**	0.55	**0.86**	**0.62**
k-means	MLM-time + ToD	0.81	0.53	0.88	0.78	**0.88**	**0.56**	0.86	**0.62**
SCAN	MLM	0.81	**0.59**	0.88	0.70	0.82	0.48	0.83	0.59
SCAN	MLM-time	0.79	0.51	0.85	0.71	0.81	0.46	0.82	0.56
DEC	MLM	0.78	0.50	0.88	0.74	0.80	0.41	0.82	0.55
DEC	MLM-time	0.79	0.51	**0.89**	0.76	0.80	0.41	0.83	0.56

Table 3: L1 (raw CASAS activities), rare classes masked from the average; $m = 5$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.70	0.46	0.88	0.65	0.21	0.21	0.60	0.44
k-means	Sensor+ToD hist.	0.70	0.37	0.77	0.34	0.70	0.38	0.72	0.36
k-means	MLM	0.77	0.50	**0.89**	**0.66**	0.28	0.26	0.65	**0.47**
k-means	MLM-time	0.72	0.41	0.88	0.66	0.27	0.27	0.62	0.44
k-means	MLM + ToD	0.71	0.32	0.80	0.35	0.73	**0.46**	0.75	0.38
k-means	MLM-time + ToD	0.75	0.40	0.76	0.39	**0.75**	0.45	**0.75**	0.42
SCAN	MLM	0.76	0.40	0.84	0.48	0.34	0.28	0.64	0.39
SCAN	MLM-time	0.75	0.49	0.88	0.62	0.27	0.23	0.63	0.45
DEC	MLM	0.77	0.46	0.88	0.61	0.19	0.22	0.61	0.43
DEC	MLM-time	**0.78**	**0.52**	**0.89**	0.62	0.27	0.26	0.65	0.47
$K = 100$ *clusters*									
k-means	Sensor hist.	0.78	0.52	**0.89**	**0.66**	0.30	0.29	0.65	0.49
k-means	Sensor+ToD hist.	**0.80**	0.57	0.87	0.63	0.75	0.53	0.81	0.57
k-means	MLM	0.79	0.57	**0.89**	0.65	0.36	0.33	0.68	0.52
k-means	MLM-time	0.79	**0.60**	**0.88**	0.65	0.30	0.29	0.66	0.51
k-means	MLM + ToD	**0.80**	0.55	**0.88**	0.63	**0.77**	**0.57**	**0.82**	**0.59**
k-means	MLM-time + ToD	0.80	0.54	0.87	0.62	0.74	0.55	0.81	0.57
SCAN	MLM	**0.80**	0.58	0.87	0.58	0.32	0.32	0.67	0.49
SCAN	MLM-time	0.79	0.53	0.86	0.56	0.32	0.33	0.66	0.47
DEC	MLM	0.76	0.51	0.88	0.61	0.29	0.27	0.65	0.47
DEC	MLM-time	0.78	0.53	**0.89**	0.63	0.30	0.29	0.66	0.48